# Kshitij Mishra, Ph.D.

Postdoctoral Associate | MBZUAI | Abu Dhabi, UAE

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## Research Profile

I study how AI agents reason, learn through interaction, and continually improve while remaining consistent, safe, auditable, and aligned with human needs.

- **Reasoning and learning:** post-training, self-improving agents, continual and multi-agent learning, and strategic interaction.
- **Consistency and introspection:** factual consistency, context attribution, and the influence of evidence on agent decisions.
- **Trustworthy AI systems:** safety, security, prompt-injection defense, adversarial evaluation, and auditing of tool-using agents.

**CAST research agenda:** Consistent, Adaptive, Strategic, and Trustworthy agents.

## Appointments and Research Experience

Aug 2025–present
**Postdoctoral Associate**
Mohamed bin Zayed University of Artificial Intelligence, Abu Dhabi, UAE.
- Research on reasoning, post-training, strategic interaction, and the safety and auditing of foundation-model agents.
- Developing learning methods using textual feedback, process-level rewards, and concept-level reasoning exploration; designing adversarial evaluations and defenses for tool-using agents.

Feb–Jun 2025
**Intern**
Quantiphi, Bangalore, India.
- Developed two process-level reward formulations to improve reasoning in small language models.

Jan–Oct 2023
**Applied Scientist Intern**
Amazon.com Services LLC, Los Angeles, USA.
- Worked on language-model editing and privacy-preserving language models, contributing to a Findings of EACL 2024 paper and a granted US patent.

Jul 2019–Aug 2025
**Doctoral Researcher**
Indian Institute of Technology Patna, India.
- Developed reinforcement-learning and reward-modeling methods for polite, empathetic, persuasive, and personalized conversational systems.

## Education

Jul 2019–Aug 2025
**Ph.D. in Computer Science and Engineering**
Indian Institute of Technology Patna, India. CPI: 8.79.
*Thesis: Inducing Politeness and Empathy In User-Oriented Conversational Systems.*

Aug 2017–Jun 2019
**M.Tech. in Computer Science and Engineering**
National Institute of Technology Arunachal Pradesh, India.
CGPA: 9.01; **Gold Medalist**.
*Thesis: Heuristic Learning for an Automated Theorem Prover.*

## Conference Publications

17 conference papers, including accepted work. **Kshitij Mishra** is shown in bold; dotted blue underlining marks first and joint-first authors.

NeurIPS '26
**PROBE: Learning to Audit Policy Compliance in Tool-Using LLM Agents**
**Kshitij Mishra**, Abhijith Sharma, Nils Lukas, Salem Lahlou.
NeurIPS 2026. **Accepted**.

EMNLP '26 Findings
**TRACER: Trace-Regularized Training Against Indirect Prompt Injection**
**Kshitij Mishra**, Salem Lahlou, Nils Lukas.
Findings of EMNLP 2026. **Accepted**. Code

ICML '26
**CoRe: Collaborative Reasoning via Cross Teaching**
**Kshitij Mishra**, Mirat Aubakirov, Martin Takac, Nils Lukas, Salem Lahlou.
ICML 2026. **Accepted**. Paper Code

EACL '26 Findings
**SD-E$^2$: Semantic Exploration for Reasoning Under Token Budgets**
**Kshitij Mishra**, Nils Lukas, Salem Lahlou.
Findings of EACL 2026. Paper Code

AAAI '25
**Breaking Barriers: A Paradigm Shift in Technology Accessibility for Individuals with Physical Disabilities**
**Kshitij Mishra**, Manisha Burja, Asif Ekbal.
AAAI 2025. Paper Code

COLING '25
**MedEx: Enhancing Medical Question-Answering with First-Order Logic based Reasoning and Knowledge Injection**
Aizan Zafar, **Kshitij Mishra**, Asif Ekbal.
COLING 2025. Paper Code

EMNLP '24
**ABLE: Personalized Disability Support with Politeness and Empathy Integration**
**Kshitij Mishra**, Manisha Burja, Asif Ekbal.
EMNLP 2024. Paper Code

EMNLP '24 Findings
**MedLogic-AQA: Enhancing Medical Question Answering with Abstractive Models Focusing on Logical Structures**
Aizan Zafar, **Kshitij Mishra**, Asif Ekbal.
Findings of EMNLP 2024. Paper Code

EACL '24 Findings
**Correcting Language Model Outputs by Editing Salient Layers**
**Kshitij Mishra**, Tamer Soliman, Anil Ramakrishna, Aram Galstyan, Anoop Kumar.
Findings of EACL 2024. Paper

EMNLP '23
**e-THERAPIST: I suggest you to cultivate a mindset of positivity and nurture uplifting thoughts**
**Kshitij Mishra**, Priyanshu Priya, Manisha Burja, Asif Ekbal.
EMNLP 2023. Paper Code

ACL '23
**PAL to Lend a Helping Hand: Towards Building an Emotion Adaptive Polite and Empathetic Counseling Conversational Agent**
**Kshitij Mishra**, Priyanshu Priya, Asif Ekbal.
ACL 2023. Paper Code

IJCAI '23
**PARTNER: A Persuasive Mental Health and Legal Counselling Dialogue System for Women and Children Crime Victims**
Priyanshu Priya, **Kshitij Mishra**, Palak Totala, Asif Ekbal.
IJCAI 2023. Paper Code

## Conference Publications (continued)

EACL '23 — **RPTCS: A Reinforced Persona-aware Topic-guiding Conversational System**
Zishan Ahmad, **Kshitij Mishra**, Asif Ekbal, Pushpak Bhattacharyya.
EACL 2023. Paper Code

AAAI '23 — **Help Me Heal: A Reinforced Polite and Empathetic Mental Health and Legal Counseling Dialogue System for Crime Victims**
**Kshitij Mishra**, Priyanshu Priya, Asif Ekbal.
AAAI 2023. Paper Code

COLING '22 — **PEPDS: A Polite and Empathetic Persuasive Dialogue System for Charity Donation**
**Kshitij Mishra**, Azlaan Mustafa Samad, Palak Totala, Asif Ekbal.
COLING 2022. Paper Code

NAACL '22 Findings — **Empathetic Persuasion: Reinforcing Empathy and Persuasiveness in Dialogue Systems**
Azlaan Mustafa Samad, **Kshitij Mishra**, Mauajama Firdaus, Asif Ekbal.
Findings of NAACL 2022. Paper

IEEE SMC '19 — **Regularized Universum twin support vector machine for classification of EEG Signal**
Deepak Gupta, Hemanga Jyoti Sarma, **Kshitij Mishra**, Mukesh Prasad.
IEEE SMC 2019. Paper

## Journal Publications

6 journal articles. Published research articles, listed by year.

IEEE/ACM TASLP '24 — **Please Donate to Save a Life: Inducing Politeness to Handle Resistance in Persuasive Dialogue Agents**
**Kshitij Mishra**, Mauajama Firdaus, Asif Ekbal.
IEEE/ACM Transactions on Audio, Speech, and Language Processing (2024), 32, 2202–2212. Paper Code

PLOS ONE '23 — **GenPADS: Reinforcing politeness in an end-to-end dialogue system**
**Kshitij Mishra**, Mauajama Firdaus, Asif Ekbal.
PLOS ONE (2023), 18(1), e0278323. Paper Code

IEEE TCSS '23 — **Predicting Politeness Variations in Goal-Oriented Conversations**
**Kshitij Mishra**, Mauajama Firdaus, Asif Ekbal.
IEEE Transactions on Computational Social Systems (2023), 10(3), 1095–1104. Paper Code

Neurocomp. '22 — **Please be polite: Towards building a politeness adaptive dialogue system for goal-oriented conversations**
**Kshitij Mishra**, Mauajama Firdaus, Asif Ekbal.
Neurocomputing (2022), 494, 242–254. Paper Code

Environ. Earth Sci. '21 — **Artificial intelligence for suspended sediment load prediction: a review**
Deepak Gupta, Barenya Bikash Hazarika, Mohanadhas Berlin, Usha Mary Sharma, **Kshitij Mishra**.
Environmental Earth Sciences (2021), 80, article 346. Paper

Research Cell '19 — **Artificial Neural Networks in Hydrology**
**Kshitij Mishra**, Deepak Gupta.
Research Cell: An International Journal of Engineering Science (2019), 31, 52–59 (Special Issue, January). Paper

## Patent

**Systems and methods for updating large language models**

**Kshitij Mishra**, Tamer Soliman, Aram Galstyan, Anoop Kumar, and Anil K Ramakrishna.

US Patent 12,456,020 B1. Granted October 28, 2025. Amazon Technologies, Inc. Patent record

## Deployed Research Systems

**Etihad Conversational Booking Agent.** Designed, developed, and deployed a conversational agent for natural-language flight search and booking assistance at MBZUAI.

**Sevak, an Indian-language chatbot.** Worked with two fellow researchers to deploy 14 multilingual conversational systems covering railways, judiciary, and healthcare during my Ph.D. at IIT Patna.

**Research software:** CoRe • SD-E2 • PEPDS • PAL • ABLE • EDiSS

## Teaching and Mentoring

**Indian Institute of Technology Patna**

Supported teaching in theory of computation, machine learning, natural language processing, and artificial intelligence. Mentored students across academic levels and contributed to research on conversational AI.

## Academic Service

2022 Organizer, GIAN workshop on *Deep Learning Techniques for Conversational Artificial Intelligence*, IIT Patna, April 13–24.

2020 Organizer, CEP course on *Deep Learning for NLP*, IIT Patna, January 10–23.

2018 Organizer, workshop on *Recent Trends in Computing and Computer Networking*, NIT Arunachal Pradesh, September 3–7.

2017 Organizer, workshop on *Recent Trends in Machine Learning, Big Data and IoT*, NIT Arunachal Pradesh, November 13–17.

## Honors and Fellowships

2019 **Prime Minister's Research Fellowship**, Government of India.

2019 **M.Tech. Gold Medal**, NIT Arunachal Pradesh.

2017 **GATE Scholarship**, Government of India.

## Technical Skills and Languages

**Programming:** Python, C++, C, LaTeX.

**Frameworks:** PyTorch, TensorFlow, scikit-learn, Hugging Face, Google Colab.

**Languages:** English and Hindi.

## Additional Training

2019 *Machine Learning for Engineering and Science Application*, IIT Madras.

2018 *Introduction to Machine Learning*, IIT Kharagpur.